Beyond Synthetic Accuracy: A Counterfactual Evaluation of Learned ADCS Prerequisite Detection

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Abstract—High classification accuracy on synthetic security graphs does not establish that a detector resolves the authorization dependencies it is intended to model. We examine this distinction for CertGraph, a heterogeneous graph attention classifier of Active Directory Certificate Services (ADCS) configurations. First, we rebuild evaluation of its seven-class generator with disjoint input manifests, validation-only checkpoint selection, and structural tabular baselines. Across three seeds, CertGraph achieves 99.71% mean macro-F1; a graph-count random forest reaches 99.90%, while a post-hoc generator-matching rule reaches 100%. Second, we construct paired positive and negative cases in an explicit, fully observed abstraction of ESC13 prerequisites. Each family changes one prerequisite and remains within one data split. CertGraph achieves 89.56% mean accuracy and classifies both variants correctly in 79.11% of test families, compared with 95% and 90% for the graph-count random forest. On remote membership interventions, all four learned baselines obtain 50% accuracy and no completely correct pairs. We connect this result to the standard locality limitation of finite-depth message passing. Finally, we measure screening followed by symbolic verification: the neural cascade checks fewer candidates but is slower than the reference query on these small graphs. The contribution is a reproducible diagnostic protocol and bounded empirical findings, rather than a new ADCS exploit detector or evidence of production superiority. Validation against independently collected Windows environments remains necessary.

Index Terms—Active Directory Certificate Services, graph neural networks, counterfactual evaluation, synthetic benchmarks, symbolic verification.

I. Introduction

ADCS security depends on interactions among certificate configuration, enrollment permissions, directory membership, and trust. The Certified Pre-Owned work documented important abuse opportunities in this infrastructure [1]. Such dependencies make graphs a natural representation, but they also create an evaluation problem: a classifier may predict a generator's label from a few correlated attributes without resolving a principal's actual prerequisites.

This paper evaluates that problem in CertGraph, a two-layer heterogeneous graph attention model developed for template classification. We ask three questions. (RQ1) Does strong performance on its existing synthetic seven-class task establish a benefit over simple structural baselines? (RQ2) Does the model distinguish paired cases whose labels differ because one prerequisite changes? (RQ3) Does learned screening improve the cost of exact verification while retaining positives? These questions concern measurable capabilities rather than whether a particular model architecture sounds appropriate for an identity graph.

Our results motivate a narrower interpretation of high synthetic scores. Simple counts nearly solve the original task. When flag-bearing safe cases are excluded from training, both the GNN and structural tabular models generalize poorly to those cases. In the new paired task, aggregate accuracy conceals a complete failure on remote membership interventions. Screening can reduce the number of exact checks yet increase total runtime, because the exact predicate is inexpensive in this setting.

The contributions are: (i) an auditable holdout protocol with namespaced generation, input-overlap checks, saved predictions, and validation-only selection; (ii) a principal-specific paired benchmark with ten interventions and explicit abstraction boundaries; (iii) evaluation under path-length and background-size changes, including family-level metrics; and (iv) a measured comparison of learned screening with direct symbolic verification. These are contributions of this artifact and study. We do not claim the first typed ADCS graph, a new graph-attention mechanism, or a general solution to ADCS exploitability. The current evidence supports an empirical benchmark study; independent operational validation is still absent.

II. Background and Related Work

A. ADCS analysis and existing tools

ESC13 concerns certificate issuance policies linked to directory groups. Under suitable enrollment and authentication prerequisites, a certificate can confer authorization associated with the linked group [2]. Its published conditions include group-link eligibility and CA trust and publication requirements. This makes it useful for studying dependency reasoning, but a Boolean graph abstraction does not implement all Windows security semantics.

Existing tooling already addresses ADCS relationships. BloodHound models ADCS entities and attack prerequisites [3]. Certipy's documentation explicitly describes ESC13 detection and distinguishes that detection from mapping downstream group privileges [4]. Consequently, a locally written flag rule or path query cannot be presented as a measured Certipy or BloodHound result. No executable comparison with those products is reported here.

B. Learning on security graphs

Graph attention assigns learned weights to neighborhood messages [5]. CertGraph applies relation-specific at-

tention to typed graphs and adds type-specific linear skip projections. Residual mechanisms and identity mappings have substantial prior art, including GCNII [6]; the use of skip projections is not, by itself, a novel contribution. Nor does adding a residual term universally guarantee a positive lower bound on a network's input Jacobian: branches can cancel, activations can have zero derivative, and projections can lose rank.

Arp et al. [7] discuss how evaluation choices can yield misleading conclusions in security machine learning. Our experiments instantiate that concern for a specific synthetic ADCS task through independent split generation, simple baselines, and interventions on prerequisites. The resulting locality statement is standard message-passing reasoning, not a new expressivity theorem.

Defensive optimization is a related but separate problem. For example, Guo et al. [8] study budgeted edge blocking on AD-style attack graphs. Our experiments neither evaluate autonomous permission changes nor establish disruption-aware remediation. These topics are excluded from the empirical contribution.

III. Tasks and Abstraction Boundaries

A. Legacy seven-class task

The existing generator assigns one template label from Safe, ESC1, ESC2, ESC3, ESC4, ESC9, and ESC13. Each sample is a small typed graph with synthetic certificate attributes and relationships. This is a mutually exclusive generator classification task, even though operational weaknesses can coexist. The term Safe means the generator's negative class; it is not a claim that a real deployment is secure.

We retain this generator to test what its reported accuracy means. A source-derived diagnostic applies a short sequence of configuration and relation checks matching its construction. This baseline is intentionally post hoc. Perfect diagnostic accuracy establishes that these generated labels are recoverable by the implemented rules on the sampled inputs; it does not validate those rules as a complete ADCS specification.

B. Principal-specific ESC13 abstraction

The second task evaluates a focal user u and template t. Its five node types are User, Group, Template, CA, and Policy. Seven forward relation types encode user membership, nested group membership, group enrollment on a template, user enrollment on a CA, template publication, template policy association, and policy-to-group linking. Reverse relations are supplied only for neural message passing, yielding fourteen computational relation types. They do not confer reverse authorization.

The benchmark operates on resolved effective-allow relationships. It assumes eligible empty universal policy groups and a pre-resolved CA trust/availability flag. It excludes raw DACL deny precedence, inheritance resolution, SID filtering, cross-forest rules, patch-specific KDC behavior, relay, enrollment-agent chains, and incomplete collection. Approval and signature requirements are treated as unmet by the focal user. Consequently, the target is a prerequisite predicate within this abstraction, not a Windows exploitability oracle.

Let $M(u)$ be groups reachable from u by one user-membership edge followed by zero or more nested-group edges. Define

$$\begin{aligned} A(t) &= \mathrm{Auth}(t) \wedge \neg\,\mathrm{Approval}(t) \\ &\quad \wedge \neg\,\mathrm{Signature}(t), && (1)\\ E(u,t) &= \exists g \in M(u) : \mathrm{Enroll}(g,t), && (2)\\ C(u,t) &= \exists c : \mathrm{Published}(t,c) \\ &\quad \wedge \mathrm{EnrollCA}(u,c) \wedge \mathrm{Trusted}(c), && (3)\\ P(t) &= \exists p, g : \mathrm{HasPolicy}(t,p) \\ &\quad \wedge \mathrm{Present}(p) \wedge \mathrm{Linked}(p,g) \\ &\quad \wedge \mathrm{HighValue}(g). && (4) \end{aligned}$$

The binary label is

$$y(u,t) = A(t) \wedge E(u,t) \wedge C(u,t) \wedge P(t). \quad (5)$$

HighValue is an input annotation of the authorization objective; its correctness is assumed rather than learned. This assumption deliberately separates resolving prerequisites from inferring the business importance of a group.

C. Differential reference checking

The primary reference uses queue-based membership traversal followed by relation intersections. A second implementation computes Boolean matrix transitive closure and evaluates the predicate independently. Both agree on every generated case in the reported paired experiment. Additional tests remove arbitrary edges and compare their answers, so agreement is not checked solely on untouched generator outputs.

These checks can expose coding errors in either implementation. They cannot detect a shared mistake in the predicate, and they do not establish correspondence with Windows behavior. The reference's 100% result is a consistency result because it defines the labels. Treating that score as an independent detection achievement would be circular.

IV. Counterfactual Construction

Each family contains a positive case and a negative variant. Two users connect to symmetric membership chains. One chain has enrollment on the target template; the other has enrollment on a distractor. Both templates share a CA and issuance policy. Background groups contain optional cycles in a disconnected component. A shared random permutation of node indices is applied to both variants. Family identifiers, variant names, mutation identifiers, and labels are stored for auditing but are not model features.

Ten intervention types are equally represented in the mixed test sets: exchange the users' chain memberships;

move the target enrollment grant to the distractor; remove focal CA enrollment; remove target publication; disable authentication; require approval; require a signature; remove the target policy association; remove the linked group's high-value annotation; or disable CA trust. Each changes one semantic prerequisite. Some require rewiring several edges, so "one intervention" does not mean exactly one edge edit. The privilege annotation intervention changes the stated target objective rather than simulating an ACL repair.

Most interventions change a directly visible attribute or count. The membership exchange is deliberately different: it preserves the template configuration and relation degree counts while changing which user can reach the enrolling group. Thus the mixed benchmark includes both easy local evidence and a controlled relational ambiguity. It is not a balanced survey of all possible ADCS configurations.

A. Splits and shifts

For each seed, training contains 300 families (600 graphs) and validation contains 100 families (200 graphs). Each of the IID, long-path, and larger-graph tests has 150 families (300 graphs). A targeted remote-membership test has 100 families (200 graphs). Families never cross splits. Training and IID depths contain zero to three group-to-group edges per chain; the long-path and remote tests contain five to eight. The larger-graph test increases background groups from 8–32 to 128–256 while retaining training-range depths.

Independent namespace strings are included in deterministic generation seeds. We also hash model-observable node tensors, edge indices, and target indices to reject exact cross-split duplicates. This is an exact representation check, not graph-isomorphism deduplication. Repeated motifs and isomorphic local neighborhoods are expected, including by design in the remote test. Consequently, IID means held-out families from the same construction, not independent organizations or novel security mechanisms.

V. Models, Selection, and Metrics

A. Compared methods

CertGraph has two heterogeneous attention layers with relation outputs summed by node type. The first layer uses four heads of width 32 (128 concatenated channels); the second uses one head of width 16. Type-specific linear skips accompany both layers. ReLU and dropout of 0.2 are used, followed by a 16-to-16-to-output classifier. There are seven outputs for the legacy task and two for the prerequisite task. Lazy parameters are initialized before constructing the optimizer.

Feature MLP uses only target template attributes. Graph-count MLP and Graph-count RF add direct structural summaries, without membership closure. On the paired task these include trusted-CA and high-value-group counts, target enrollment/publication/policy counts, focal CA enrollment count, policy-link count, and nested-membership edge count. These engineered features give the tabular methods useful global summaries; they are not identical to the GNN's finite receptive field. Their purpose is to test whether a simpler representation suffices, not to claim information-equivalent architectures.

The MLP has two hidden layers of width 32, with ReLU and dropout of 0.2. The random forest has 200 trees with otherwise library defaults. The paired reference receives the complete abstract graph. For the legacy task, we additionally retain the historical direct-edge rules under that descriptive name and add the post-hoc generator diagnostic. Neither is an external-tool execution.

B. Training and holdouts

All runs use seeds 11, 29, and 47. The GNN uses Adam, learning rate 0.005, weight decay 10^{-4}, batches of 64 graphs, at most 80 epochs, and patience 15. Checkpoints minimize validation negative log-likelihood; test scores never select an epoch or seed. MLPs use Adam at 0.01, the same weight decay, at most 250 full-batch epochs, and patience 35. Scaling statistics are fitted on training inputs only, with each standard deviation floored at one. Hyperparameters are fixed across reported runs; there is no exhaustive architecture search.

Each legacy run contains 700 training, 210 validation, and 350 test graphs. In the mixed-training condition, all splits contain generator hard negatives. In the no-HN-training condition, training and validation exclude them but testing includes them. The two conditions have separately generated datasets, including test sets. Their difference is descriptive rather than a matched-sample causal estimate of the training intervention.

C. Reported quantities

For seven classes we report macro-F1, with the label set fixed across runs. We separately report specificity for flag-bearing safe examples. For binary cases we save accuracy, macro-F1, recall, specificity, precision, average precision, and confusion matrices. A pair is completely correct only if both its positive and negative variant are classified correctly:

$$B = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\hat{y}_i^+ = 1 \wedge \hat{y}_i^- = 0]. \tag{6}$$

The family, not its two correlated rows, is the resampling unit for saved percentile bootstrap intervals (2,000 resamples). Tables summarize three-seed means and sample standard deviations; these are not population confidence intervals. Three runs and restricted motifs cannot support broad statistical generalization. Zero observed variance can arise because all seeds reproduce the same systematic error.

VI. Results

A. RQ1: What does legacy accuracy establish?

Table I shows that CertGraph averages 99.71% macro-F1 with mixed training, while the count MLP averages

TABLE I
Seven-class synthetic holdouts: macro-F1 (percent), mean ± sample standard deviation over three seeds. Each test has 350 graphs. The generator diagnostic is post hoc.

Method	Mixed training	No-HN training
CertGraph	99.71 ± 0.50	87.82 ± 1.20
Feature MLP	86.21 ± 2.24	86.88 ± 1.89
Graph-count MLP	99.81 ± 0.16	87.99 ± 0.96
Graph-count RF	99.90 ± 0.16	88.27 ± 1.19
Generator diagnostic	100.00 ± 0.00	100.00 ± 0.00
Legacy direct-edge rules	90.99 ± 0.59	94.32 ± 0.68

TABLE II
Specificity on flag-bearing safe examples in the seven-class test (percent). These are a diagnostic subset, not an independently collected negative population.

Method	Mixed training	No-HN training
CertGraph	97.62 ± 4.12	5.16 ± 3.98
Feature MLP	0.00 ± 0.00	0.00 ± 0.00
Graph-count MLP	100.00 ± 0.00	5.69 ± 9.86
Graph-count RF	100.00 ± 0.00	7.32 ± 12.67

99.81% and the count random forest 99.90%. The source-derived diagnostic obtains 100% in both conditions. These observations do not establish a GNN advantage over simple structural features. They also do not prove equivalence of the learned methods; no equivalence margin or adequately powered equivalence test was specified.

Without hard negatives in training, mean GNN macro-F1 falls to 87.82%. Table II isolates the affected subset: mean GNN specificity on flag-bearing safe cases is 5.16%, versus 97.62% under mixed training. The feature-only MLP has zero specificity in both conditions. The count methods recover the mixed-training negatives but also fail on most negatives excluded from training. This supports a data-support limitation shared by the learned baselines, rather than a unique failure of neural graph processing or evidence that one previously deployed checkpoint failed.

B. RQ2: Which prerequisites are distinguished?

CertGraph averages 89.56% case accuracy on IID families, but only 79.11% complete-pair correctness. The count MLP and forest obtain 95% and 90%, respectively (Tables III and IV). The feature MLP reaches 65% accuracy by resolving the three template-attribute interventions. Its 30% pair correctness more directly exposes the limited set of prerequisites it distinguishes.

The per-intervention records show 50% GNN accuracy on both membership exchange and template-enrollment removal in all three IID runs. Eight intervention types are perfectly classified in two runs; the third also makes errors on the privilege-annotation intervention. This diagnoses the observed errors without attributing every failure to architectural locality.

The long-path mixture does not materially lower aggregate accuracy: many interventions remain locally visible even when the membership chain is longer. Thus a high score on a mixture of long-path graphs is insufficient evidence of long-range reasoning. On the targeted remote-membership set, every learned method obtains 50% accuracy and 0% complete-pair correctness. All predict the positive class for both variants under the default decision rule. Figure 1 contrasts those metrics with the mixed tests.

Increasing only disconnected background size reduces the count MLP's mean accuracy to 67.56%, while the count forest remains at 95%. The GNN averages 89.67%. The background edge count is included in the engineered feature vector, so this shift tests extrapolation of that representation as well as the classifier. It is not evidence that all tabular learners fail on larger directories, nor a scalability test on enterprise data.

C. Locality explains a controlled failure

Proposition 1 (Standard message-passing locality). Consider a deterministic L-layer message-passing model whose updates use only a node's current representation and typed incoming-neighbor messages, with no global pooling or global structural encoding. If two targets have isomorphic feature- and relation-preserving depth-L computation neighborhoods, their output logits are equal. Nodewise residual projections do not change this statement.

Proof. Corresponding depth-zero representations are equal by assumption. If corresponding representations at depth k are equal, each corresponding node receives the same multiset of typed messages at the next layer. Shared update and aggregation functions therefore produce equal representations. Nodewise projections preserve equality. Induction to depth L, followed by a shared classifier, gives equal logits. □

For paired opposite labels satisfying the proposition, the same deterministic decision cannot classify both correctly. Its case accuracy over that pair is exactly one half. The remote membership construction supplies such pairs for the evaluated two-layer model; a regression test verifies equal evaluation logits on a representative pair before training. This is a designed limitation of the available representation, not a universal impossibility result for GNNs. Greater depth, principal-conditioned path summaries, global mechanisms, or precomputed closure change the assumptions and require separate evaluation.

D. RQ3: Does screening accelerate verification?

For each learned method, we set the screening threshold to the minimum positive validation score, retaining all observed validation positives. This is a permissive empirical operating point, not a guarantee on unseen positives. The final decision is the screen decision conjoined with the exact reference predicate. A cheap comparison first tests the necessary template flags and then runs the same reference.

On IID tests, CertGraph screening passes 55% of candidates to verification and retains all observed positives. Its mean corpus time is 407.31 ms, compared with 2.89 ms for reference-only evaluation (Table V). Count-MLP screening

TABLE III
Paired prerequisite benchmark: accuracy (percent), mean ± sample standard deviation over three seeds. IID, long-path, and larger-graph sets each contain 150 families per seed; remote membership contains 100. The reference also defines the labels.

Method	IID	Long paths	Larger graphs	Remote membership
CertGraph	89.56 ± 0.77	90.00 ± 0.00	89.67 ± 0.58	50.00 ± 0.00
Feature MLP	65.00 ± 0.00	65.00 ± 0.00	65.00 ± 0.00	50.00 ± 0.00
Graph-count MLP	95.00 ± 0.00	95.00 ± 0.00	67.56 ± 15.56	50.00 ± 0.00
Graph-count RF	95.00 ± 0.00	95.00 ± 0.00	95.00 ± 0.00	50.00 ± 0.00
Symbolic reference	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00

TABLE IV
Families with both variants classified correctly (percent), mean ± sample standard deviation. Each pair contains one positive and one negative.

Method	IID	Long paths	Larger graphs	Remote membership
CertGraph	79.11 ± 1.54	80.00 ± 0.00	79.33 ± 1.15	0.00 ± 0.00
Feature MLP	30.00 ± 0.00	30.00 ± 0.00	30.00 ± 0.00	0.00 ± 0.00
Graph-count MLP	90.00 ± 0.00	90.00 ± 0.00	35.11 ± 31.13	0.00 ± 0.00
Graph-count RF	90.00 ± 0.00	90.00 ± 0.00	90.00 ± 0.00	0.00 ± 0.00
Symbolic reference	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00

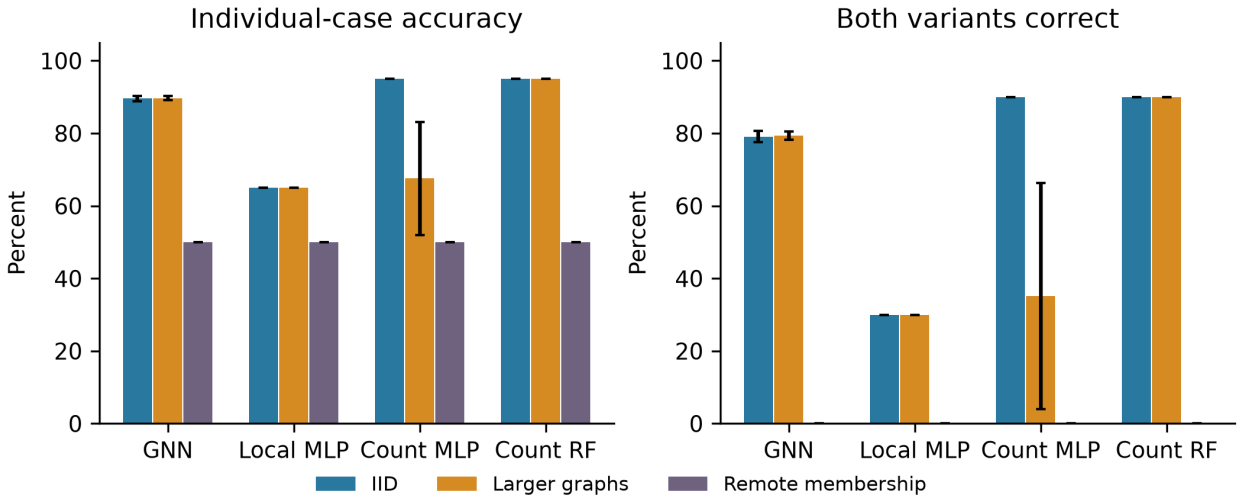


Fig. 1. Observed accuracy and complete-pair correctness. Bars are means over three seeds, with sample standard deviation error bars. The remote test consists solely of degree-preserving membership exchanges. A 50% case score there corresponds to zero completely correct pairs.

also checks 55%, but takes 5.15 ms. The RF cascade has mean recall 99.78%, showing that retaining every validation positive does not ensure retention on a test set. The cheap necessary-flags cascade checks 85% and takes 3.64 ms; the reference already short-circuits on those flags, explaining why an external prefilter need not help.

These timings are in-memory microbenchmarks over 300 small graph cases. They include model batching or feature extraction and selected verification, but exclude raw collector ingestion, network transfer, and persistent graph-database indexing. Each callable is warmed up; we use the median of five reference/flag repetitions and three learned-pipeline repetitions per seed. CPU contention and the small number of repeats limit precise performance inference. The data nevertheless provide no measured acceleration benefit from the tested GNN cascade in this setting.

If S is the screening decision and V is the verifier decision, the cascade accepts only when $S \wedge V$. Consequently,

$$\Pr(S \wedge V = 1 \mid Y = 1) \leq \Pr(S = 1 \mid Y = 1). \qquad (7)$$

An exact verifier cannot recover a positive that screening discards. Zero false positives follow by construction only when the verifier is sound for the target labels; here that statement is restricted to our prerequisite predicate. It is not a zero-error guarantee for operational ADCS.

VII. Validity, Reproducibility, and Scope

A. Internal validity

Saved manifests and namespace separation address exact train–test overlap. Validation-only selection avoids choosing checkpoints by test performance. Checkpoints, per-case scores, confusion matrices, training histories, source hashes, and package versions are retained. The

TABLE V
IID screening followed by exact checking within the abstraction. Corpus latency covers 300 graphs in memory. Values are mean ± sample standard deviation of per-seed medians; they are not deployment latency.

Screen	Recall (%)	Checked (%)	Time (ms)
CertGraph	100.00 ± 0.00	55.00 ± 0.00	407.31 ± 19.88
Feature MLP	100.00 ± 0.00	85.00 ± 0.00	3.68 ± 1.14
Graph-count MLP	100.00 ± 0.00	55.00 ± 0.00	5.15 ± 0.97
Graph-count RF	99.78 ± 0.38	54.89 ± 0.19	52.02 ± 6.28
Reference only	100.00	100.00	2.89 ± 0.94
Flags + reference	100.00	85.00	3.64 ± 1.67

reference differential tests address implementation consistency. These controls do not remove specification bias, generator shortcuts, or the information differences among local, count-based, and full-graph methods.

The interventions were designed after reviewing the legacy system, so this is an exploratory diagnostic study rather than a preregistered confirmatory experiment. Model settings were fixed for the reported runs, and all completed default seeds are included. Pair-preserving construction is stronger than independently drawing positive and negative graphs for the stated diagnostic question, but the restricted family templates make many cases much simpler than actual environments.

B. External validity

No result in this paper is independently labeled by certificate enrollment and authentication outcomes on Windows. Policy-group eligibility and trust are assumed inputs, and effective enrollment is already resolved. The generators do not model all authorization conflicts, collection errors, overlapping ESC conditions, or realistic class prevalence. In particular, "negative" means that this one predicate is false, not that the environment has no exploitable path.

Existing GOAD-based artifacts in the workspace combine collected directory topology with synthetic certificate configurations. Such topology scaffolds do not constitute independent ADCS validation and are not counted as additional deployment environments here. Likewise, a locally implemented tiered generator inspired by another project is not an evaluation of that upstream generator. Claims about enterprise robustness require separate data provenance and independently checked labels.

C. Reproduction artifact

The revision stores six legacy runs and three paired-task runs under results/publication_validation. A table builder reads completed result files, verifies the expected seed/condition coverage, and produces the manuscript tables and plot. Predictions retain input hashes or family identifiers for reanalysis. Paired cases are distributed as compressed JSON alongside tensor-model checkpoints. Source fingerprints and package versions record the execution environment; the reported runs use Python 3.11.11, PyTorch 2.12.0 CPU, and PyTorch Geometric 2.8.0, with two PyTorch threads.

The accompanying protocol specifies how independent Windows cases and real tool runs should be recorded. Those records are currently absent. Thus the artifact is reproducible synthetic evidence with a concrete external-validation plan, not a completed production benchmark. Historical outputs are preserved separately so that corrections do not erase the provenance of earlier claims.

VIII. Implications and Future Validation

The results suggest three evaluation requirements for learned authorization analysis. First, compare against direct rules and inexpensive structural features before attributing accuracy to message passing. Second, report prerequisite-specific and complete-pair performance: mixture accuracy can obscure failure on the relational dependency motivating the model. Third, measure the full screening-plus-verification path and compare it with a short-circuiting exact predicate. Fewer candidates are not themselves evidence of lower cost.

A stronger follow-up should collect independent ADCS configurations, freeze labels before model evaluation, and retain evidence of effective permissions, published templates, trust, policy links, and observed outcomes. The same authorized snapshots should support version-pinned Certipy and BloodHound evaluations, with clearly defined output-to-task mappings. Splits should separate environment and time, rather than only generator seeds. Multiple prerequisite interventions should include inheritance conflicts, missing data, alternate paths, and overlapping weaknesses.

Where a fully observed predicate is cheap, this study supports retaining exact verification. A useful learned extension would need a different demonstrated advantage, such as calibrated ranking under incomplete observations or amortized work at measured scale. Such claims require new experiments; the present data do not establish them. Autonomous remediation additionally requires valid action semantics, business constraints, and outcome measurements, and remains outside this paper.

IX. Conclusion

The reconstructed evaluation distinguishes synthetic label prediction from prerequisite reasoning. CertGraph's near-perfect legacy score is matched by simpler structural baselines, while paired interventions reveal unresolved membership dependencies. A learned screening cascade

reduces checks but provides no runtime advantage over the tested exact reference on small in-memory graphs. The resulting artifact provides auditable splits, paired cases, differential checks, and measured failure modes. Its findings are bounded by the abstraction and await independent Windows validation before supporting operational detection claims.

References

[1] W. Schroeder and L. Christensen, "Certified pre-owned: Abusing active directory certificate services," SpecterOps, Tech. Rep., 2021. [Online]. Available: https://specterops.io/wp-content/uploads/sites/3/2022/06/Certified_Pre-Owned.pdf

[2] J. B. Knudsen, "ADCS ESC13 abuse technique," SpecterOps technical article, Feb. 2024. [Online]. Available: https://specterops.io/blog/2024/02/14/adcs-esc13-abuse-technique/

[3] SpecterOps, "ADCS attack paths in BloodHound—part 1," Jan. 2024. [Online]. Available: https://specterops.io/blog/2024/01/24/adcs-attack-paths-in-bloodhound-part-1/

[4] O. Lyak, "Certipy documentation: Introduction," 2026, accessed October 14, 2026. [Online]. Available: https://github.com/ly4k/Certipy/wiki/01-%E2%80%90-Introduction

[5] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph attention networks," in International Conference on Learning Representations, 2018. [Online]. Available: https://arxiv.org/abs/1710.10903

[6] M. Chen, Z. Wei, Z. Huang, B. Ding, and Y. Li, "Simple and deep graph convolutional networks," in Proceedings of the 37th International Conference on Machine Learning, ser. Proceedings of Machine Learning Research, vol. 119, 2020, pp. 1725–1735. [Online]. Available: https://proceedings.mlr.press/v119/chen20v.html

[7] D. Arp, E. Quiring, F. Pendlebury, A. Warnecke, F. Pierazzi, C. Wressnegger, L. Cavallaro, and K. Rieck, "Dos and don'ts of machine learning in computer security," in 31st USENIX Security Symposium, 2022, pp. 3971–3988. [Online]. Available: https://www.usenix.org/conference/usenixsecurity22/presentation/arp

[8] M. Guo, M. Ward, A. Neumann, F. Neumann, and H. Nguyen, "Scalable edge blocking algorithms for defending active directory style attack graphs," in Proceedings of the AAAI Conference on Artificial Intelligence, vol. 37, no. 5, 2023, pp. 5649–5656.